EE 451: Communications Systems — Midterm Exam 1 Formula Sheet

Fourier Transform Pairs:

Time Domain	Frequency Domain
$\delta(t)$	1
1	$\delta(f)$
$e^{-at}u(t),\ a>0$	$\frac{1}{a+j2\pi f}$
$\text{rect}(t/T)$	$T \cdot \text{sinc}(fT)$
$\cos(2\pi f_0 t)$	$\frac{1}{2}[\delta(f-f_0)+\delta(f+f_0)]$
$\sin(2\pi f_0 t)$	$\frac{j}{2}[\delta(f+f_0)-\delta(f-f_0)]$

Fourier Transform Properties:

Property	Time Domain	Frequency Domain
Linearity	$ax_1(t)+bx_2(t)$	$aX_1(f)+bX_2(f)$
Time shift	$x(t-t_0)$	$X(f)e^{-j2\pi f t_0}$
Frequency shift	$x(t)e^{j2\pi f_0 t}$	$X(f-f_0)$
Scaling	$x(at)$	$\frac{1}{\lvert a\rvert}X(f/a)$
Convolution	$x(t)*h(t)$	$X(f)\cdot H(f)$
Modulation	$x(t)\cos(2\pi f_0 t)$	$\frac{1}{2}[X(f-f_0)+X(f+f_0)]$

Parseval's Theorem:

$$E = \int_{-\infty}^{\infty} |x(t)|^2\,dt = \int_{-\infty}^{\infty} |X(f)|^2\,df$$

Useful Integral:

$$\int_{-\infty}^{\infty} \text{sinc}^2(x)\,dx = 1$$

Trigonometric Identities:

Identity	Formula
Product	$\cos(A)\cos(B) = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$
Half-angle	$\cos^2(A) = \frac{1}{2}[1 + \cos(2A)]$

AM Formulas:

Type	Formula
Full AM	$s(t) = A_c[1 + \mu\, m_n(t)]\cos(2\pi f_c t)$
Total power (general)	$P_{\text{total}} = \frac{A_c^2}{2R}\left(1 + \mu^2\langle m_n^2(t)\rangle\right)$
Total power (single tone)	$P_{\text{total}} = \frac{A_c^2}{2R}\left(1 + \frac{\mu^2}{2}\right)$ since $\langle\cos^2\rangle = \frac{1}{2}$
Efficiency	$\eta = \frac{\mu^2\langle m_n^2(t)\rangle}{1+\mu^2\langle m_n^2(t)\rangle}$
DSB-SC	$s(t) = A_c\, m(t)\cos(2\pi f_c t)$
SSB (USB)	$s(t) = \frac{1}{2}A_c[m(t)\cos(2\pi f_c t) - \hat{m}(t)\sin(2\pi f_c t)]$

Receiver and Other:

Name	Formula
Image frequency (high-side injection)	$f_{\text{image}} = f_{\text{RF}} + 2f_{\text{IF}}$
Filter quality factor	$Q = f_{\text{center}}/B_{\text{channel}}$
sinc function (Haykin Eq. 2.9)	$\text{sinc}(x) = \sin(\pi x)/(\pi x)$
Euler's formula	$e^{j\theta} = \cos(\theta) + j\sin(\theta)$

Reference: Standard AM broadcast channel bandwidth in the Americas: $B_{\text{channel}} = 10$ kHz.